

1. Check whether a number is positive or negative

```
2. package Numbers;
3.
4. import java.util.Scanner;
5.
6. public class Positive_negative {
7.
8.     public static void main(String[] args) {
9.         // TODO Auto-generated method stub
10.
11.         Scanner sc = new Scanner(System.in);
12.         System.out.print("Enter number: ");
13.         int num=sc.nextInt();
14.
15.         if(num<0)
16.             System.out.println("Number is negative");
17.         else
18.             System.out.println("Number is positive");
19.
20.
21.     }
22.
23. }
```

Output:

```
24. Enter number: 5
25. Number is positive
```

2. Check whether a number is even or odd

```
package Control_Statements;

import java.util.Scanner;

public class Even_odd_check {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num=sc.nextInt();
        if(num%2 == 0) {
            System.out.print(num+" is even");
        }
        else
            System.out.print(num+ " is odd");

    }

}
```

Output:

```
Enter number: 65
65 is odd
```

3. Find the largest of two numbers

```
package Control_Statements;

import java.util.Scanner;

public class FindLargest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter two numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();

        if (a > b) System.out.println(a + " is larger.");
        else if (b > a) System.out.println(b + " is larger.");
        else System.out.println("Numbers are equal.");
        sc.close();
    }
}
```

4. Find the largest of three numbers

```
package Control_Statements;

import java.util.Scanner;

public class Largest_number {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter 1st number: ");
        int num1=sc.nextInt();
        System.out.print("Enter 2nd number: ");
        int num2=sc.nextInt();
        System.out.print("Enter 3rd number: ");
        int num3=sc.nextInt();

        if(num1>num2 && num1>num3)
            System.out.print("First number is the greatest");
        else if(num2>num3 && num2>num1)
            System.out.print("Second number is the greatest");
        else
            System.out.print("Third number is the greatest");

    }

}
```

Output:

Enter 1st number: 56

Enter 2nd number: 45
Enter 3rd number: 98
Third number is the greatest

5. Check whether a year is a leap year

```
package Numbers;

import java.util.Scanner;

public class Leap_year {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter year: ");
        int year=sc.nextInt();

        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
            System.out.println("Leap Year");
        } else {
            System.out.println("Not a Leap Year");
        }

    }

}
```

Output:

Enter year: 2042
Not a Leap Year

6. Check whether a character is a vowel or consonant

```
package Control_Statements;

import java.util.Scanner;

public class Vowels {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter character: ");
        char ch = sc.next().charAt(0);
        char temp=ch;
        ch = Character.toLowerCase(ch);

        if(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
            System.out.println("Character " + temp + " is vowel");
        else System.out.println("Character " + temp + " is consonant");
    }

}
```

```
}  
  
}
```

Output:

Enter character: T
Character T is consonant

7. Check eligibility for voting based on age

```
package Numbers;  
  
import java.util.Scanner;  
  
public class Voter {  
  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter age: ");  
        int age=sc.nextInt();  
  
        if(age>=18) {  
            System.out.print("You are eligible to vote");  
        }  
        else System.out.print("You are not eligible to vote");  
  
    }  
  
}
```

Output:

Enter age: 54
You are eligible to vote

8. Check whether a number is divisible by 5 and 11

```
package Numbers;  
  
import java.util.Scanner;  
  
public class Divisibility {  
  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter number: ");  
        int num=sc.nextInt();  
  
        if (num % 5 == 0 && num % 11 == 0) {  
            System.out.println(num + " is divisible by both 5 and 11");  
        } else {  
            System.out.println(num + " is not divisible by both 5 and 11");  
        }  
    }  
}
```

```

    }
}

}

```

Output:

Enter number: 495
495 is divisible by both 5 and 11

9. Find grade based on marks (if-else-if ladder)

```

package Control_Statements;

import java.util.Scanner;

public class Grade_system {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter marks: ");
        int marks = sc.nextInt();

        if (marks >= 90 && marks <= 100) {
            System.out.println("Grade A");
        } else if (marks >= 80) {
            System.out.println("Grade B");
        } else if (marks >= 70) {
            System.out.println("Grade C");
        } else if (marks >= 60) {
            System.out.println("Grade D");
        } else if (marks >= 45) {
            System.out.println("Grade E");
        } else if (marks >= 0) {
            System.out.println("Fail");
        } else {
            System.out.println("Invalid marks");
        }

        sc.close();

    }

}

```

Output:

Enter marks: 89
Grade B

10. Check whether a number is zero, positive, or negative

```

package Numbers;

import java.util.Scanner;

public class Positive_negative {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number: ");
        int num=sc.nextInt();

        if(num<0)
            System.out.println("Number is negative");
        else
            System.out.println("Number is positive");

    }

}

```

Output:

```

Enter number: 54
Number is positive

```

11. Find the smallest of three numbers

```

package Numbers;

import java.util.Scanner;

public class Smallest_number {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter 1st number: ");
        int num1=sc.nextInt();
        System.out.print("Enter 2nd number: ");
        int num2=sc.nextInt();
        System.out.print("Enter 3rd number: ");
        int num3=sc.nextInt();

        if(num1<num2 && num1<num3)
            System.out.print("First number is the smallest");
        else if(num2<num3 && num2<num1)
            System.out.print("Second number is the smallest");
        else
            System.out.print("Third number is the smallest");
    }

}

```

```
}
```

```
}
```

Output:

Enter 1st number: 54

Enter 2nd number: 16

Enter 3rd number: 90

Second number is the smallest

12. Check whether a person is eligible for a loan (age + salary condition)

```
package Control_Statements;
```

```
import java.util.Scanner;
```

```
public class Loan_approve {
```

```
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter age: ");  
        int age = sc.nextInt();  
        System.out.print("Enter salary: ");  
        int salary = sc.nextInt();
```

```
        if (age >= 21 && salary >= 25000) {  
            System.out.println("Eligible for loan");  
        } else {  
            System.out.println("Not eligible for loan");  
        }  
    }
```

```
}
```

```
}
```

Output:

Enter age: 24

Enter salary: 30000

Eligible for loan

13. Check whether a number is divisible by 3 or 7

```
package Numbers;
```

```
import java.util.Scanner;
```

```
public class Divisible_three {
```

```
    public static void main(String[] args) {  
        // TODO Auto-generated method stub
```

```

Scanner sc = new Scanner(System.in);
System.out.print("Enter number: ");
int num=sc.nextInt();

    if (num % 3 != 0 && num % 7 != 0) {
        System.out.println(num + " is not divisible by 3 nor 7");
    } else if(num % 3 == 0 || num % 7 == 0){

        System.out.println(num + " is divisible by 3 or 7");
    }

}

}

```

Output:

```

Enter number: 43
43 is not divisible by 3 nor 7

```

14. Simple calculator using switch case

```

package Control_Statements;

import java.util.Scanner;

public class Calculator {

    public static void main(String[] args) {
        // TODO Auto-generated method stub

        Scanner sc =new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();
        System.out.print("Enter second number: ");
        int b = sc.nextInt();
        System.out.print("Enter operator: ");
        char op = sc.next().charAt(0);

        switch(op) {
            case '+': System.out.println("Result: " + (a + b));
                       break;
            case '-': System.out.println("Result: " + (a - b));
                       break;
            case '*': System.out.println("Result: " + (a * b));
                       break;
            case '/': System.out.println("Result: " + (double)a / b);
                       break;

            default: System.out.print("Invalid");
        }
    }
}

```



```

    }

}

}
Output:
Enter first number: 56
Enter second number: 72
Enter operator: /
Result: 0.7777777777777778

```

15. Display day of week using switch case

```

package Control_Statements;

import java.util.Scanner;

public class Day_week {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Day of the week: ");
        int day = sc.nextInt();

        switch(day) {
            case 1:
                System.out.println("Monday");
                break;
            case 2:
                System.out.println("Tuesday");
                break;
            case 3:
                System.out.println("Wednesday");
                break;
            case 4:
                System.out.println("Thursday");
                break;
            case 5:
                System.out.println("Friday");
                break;
            case 6:
                System.out.println("Saturday");
                break;
            case 7:
                System.out.println("Sunday");
                break;
            default:
                System.out.println("Invalid input");
        }
    }
}

```

}

}

}

Output:

Enter Day of the week: 3

Wednesday